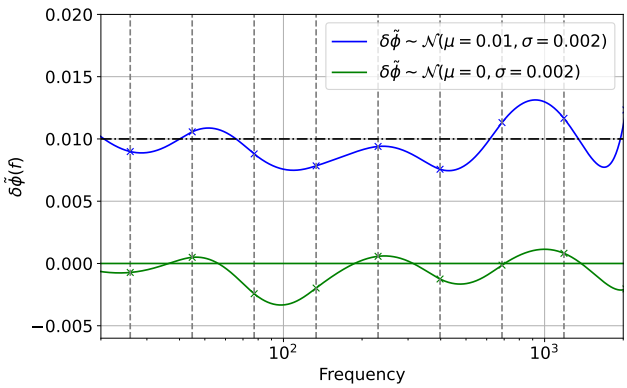


FIG. 7. We show a realization of cubic spline curves with ten frequency nodal points (vertical dashed curves) for $\delta\tilde{\phi}_i$ taken from two normal distributions: one with zero mean (green curve) and the another with non-zero mean $\mu = 0.01$ (blue curve). The points marked with a cross (x) are the realization of the normal distribution. In reality, these curves will represent a general function of frequency that captures the waveform systematics or deviation from the 'true' waveform model.

parametrization, we can account for the systematic errors introduced in the reference waveform. We can also recover the injected modification in the $\delta\phi$ parameter with correct parametrization. However, when we modify the reference waveform with rel-phase parametrization, the PE recovery with abs-phase parametrization fails to correct the observed bias in $\mathcal{M}^z$. This is because the relative-phase parametrization can accommodate more significant deviations in the reference waveform than the absolute-phase parametrization for the same range of priors in $\delta\tilde{\phi}$, as previously discussed.

B. A more realistic modification using cubic splines

Within the framework of WF-Error parametrizations, a waveform model $h_{\rm ref}$ will deviate from the 'true' model described by general functions of frequency: $\delta\tilde{A}(f), \delta\tilde{\phi}(f)$. We use cubic splines to generate them at specific frequency nodal points. In this subsection, we introduce a more general type of modification to our reference GW150914-like signal within the framework of both the parametrizations:

$$\delta\tilde{A}, \delta\tilde{\phi} \sim \mathcal{N}(\mu = 0, \sigma = 0.002) \quad (20)$$

$$\delta\tilde{A}, \delta\tilde{\phi} \sim \mathcal{N}(\mu = 0.01, \sigma = 0.002) \quad (21)$$

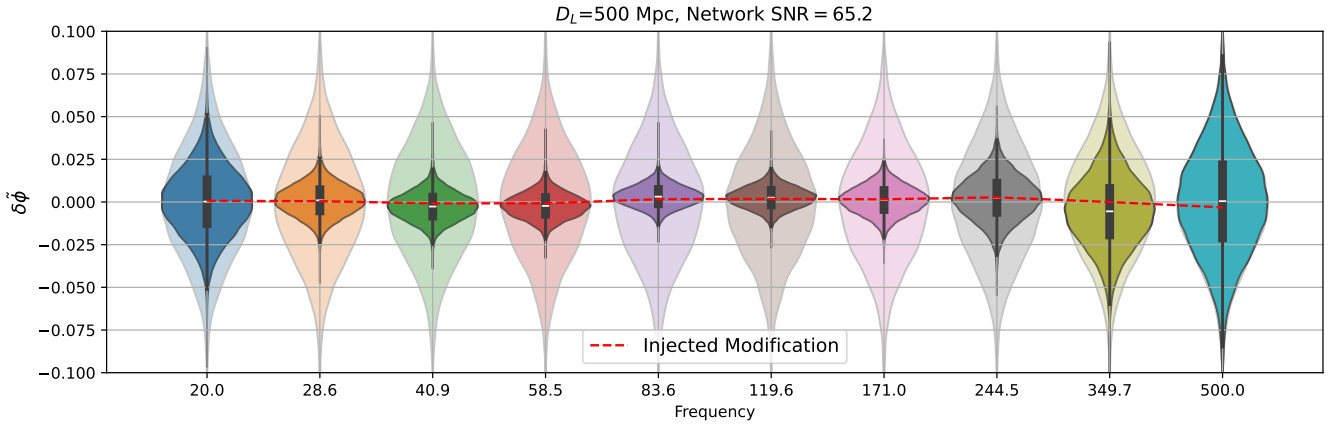
Figure 7 shows examples of cubic spline curves generated from the abovementioned distributions. We use a realization of $(\delta\tilde{A}_i, \delta\tilde{\phi}_i)$ at nodal frequency points generated from the distributions described by the equations (20) and (21).

In Figure 8, we show the recovery of $\delta\tilde{\phi}$ parameter at each frequency nodal point for the modified injection generated using rel-phase parametrization. We demonstrate that the WF-Error parametrization framework can recover the $\delta\tilde{\phi}$ curves used to modify the reference waveform model. For a relatively louder signal, corresponding to the source located at $D_L = 100$ Mpc, we observe that the posterior samples at nodal points align well with the injected values of $\delta\tilde{\phi}$. These examples demonstrate that if the reference waveform model deviates from reality, which is described by rel-phase parametrization, the framework presented here has the potential to account for that. In other words, in the absence of true knowledge of WF-Error priors, we can use a data-driven approach to determine if there is a deviation from the reference waveform model, especially for loud signals.

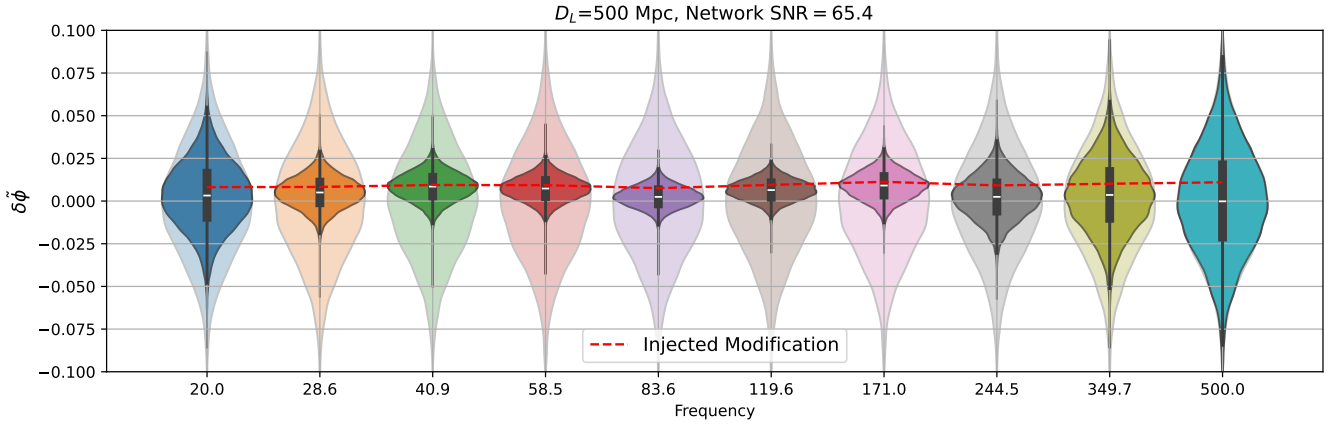
We now focus on the other parameters, such as chirp mass $\mathcal{M}$. In Figure 9, we present various combinations of injections and PE recovery. Our findings can be summarized as follows:

- When the injection uses the reference waveform model $h_{\rm ref}$ and PE recovery is performed with the same model, we achieve the expected recovery with no bias.
- If we enable the waveform error parametrizations (rel-phase or abs-phase) while using the correct recovery model, we observe a broadening of the posterior samples as anticipated. The broadening of the posterior distribution is more pronounced with the rel-phase parametrization than the abs-phase parametrization.
- In the scenario when the injection is performed by modifying the reference signal and we use the reference signal to recover the source parameters, we get bias in the chirp mass. As expected, the bias is more prominent for a large SNR signal.
- In the case of modified injections, using WF-Error parametrization in conjunction with the $h_{\rm ref}$ recovery model broadens the posteriors. Additionally, employing the correct parametrization leads to a correction in bias. The bias introduced by rel-phase parametrization can only be corrected using rel-phase parametrization. The abs-phase parametrization does not correct the biases introduced by rel-phase parametrization.
- For the abs-phase parametrization, the moderate deviation introduced by the cubic spline curves generated from the abovementioned distribution does not introduce significant bias for a GW150914-like signal with advanced LIGO detector sensitivity. In order to see any significant bias for abs-phase parametrization, we either need a comparatively large deviation in phase or a very high SNR, such as in the case of 3G detectors.

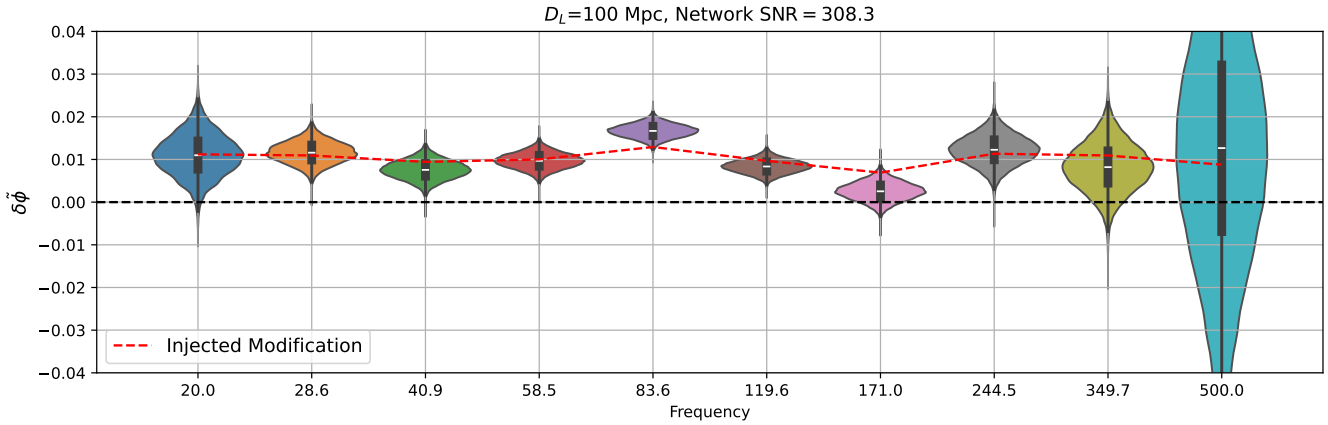
Through these simulations, we conclude that if the 'true' waveform model deviates from the reference waveform



(a)



(b)



(c)

FIG. 8. The recovery of WF-Error parameter $\delta\tilde{\phi}_i$ for rel-phase parametrization is shown for each frequency nodal point used in PE. The light violin plot represents the prior distribution for each $\delta\tilde{\phi}_i$, and the dark-shaded region represents the one-dimensional marginalized posterior samples. The last frequency nodal point at f=500 Hz returns the prior because the signal merges before this frequency. The constraints are also weaker at $f_i = 20$ Hz because this frequency bin's signal-to-noise ratio is comparatively low. a) The modification to the reference waveform is applied using the rel-phase parametrization with a cubic spline realization generated from the distribution $\delta\tilde{A}_i, \delta\tilde{\phi}_i \sim \mathcal{N}(\mu = 0, \sigma = 0.002)$. Panel b) **and c)** represent the modification from the distribution $\delta\tilde{A}_i, \delta\tilde{\phi}_i \sim \mathcal{N}(\mu = 0.01, \sigma = 0.002)$. The network SNR and luminosity distance of the injected (GW150914-like) signal are shown at the top of each panel. For these PE runs, we fix the parameters RA, dec, and polarization to the injected values.

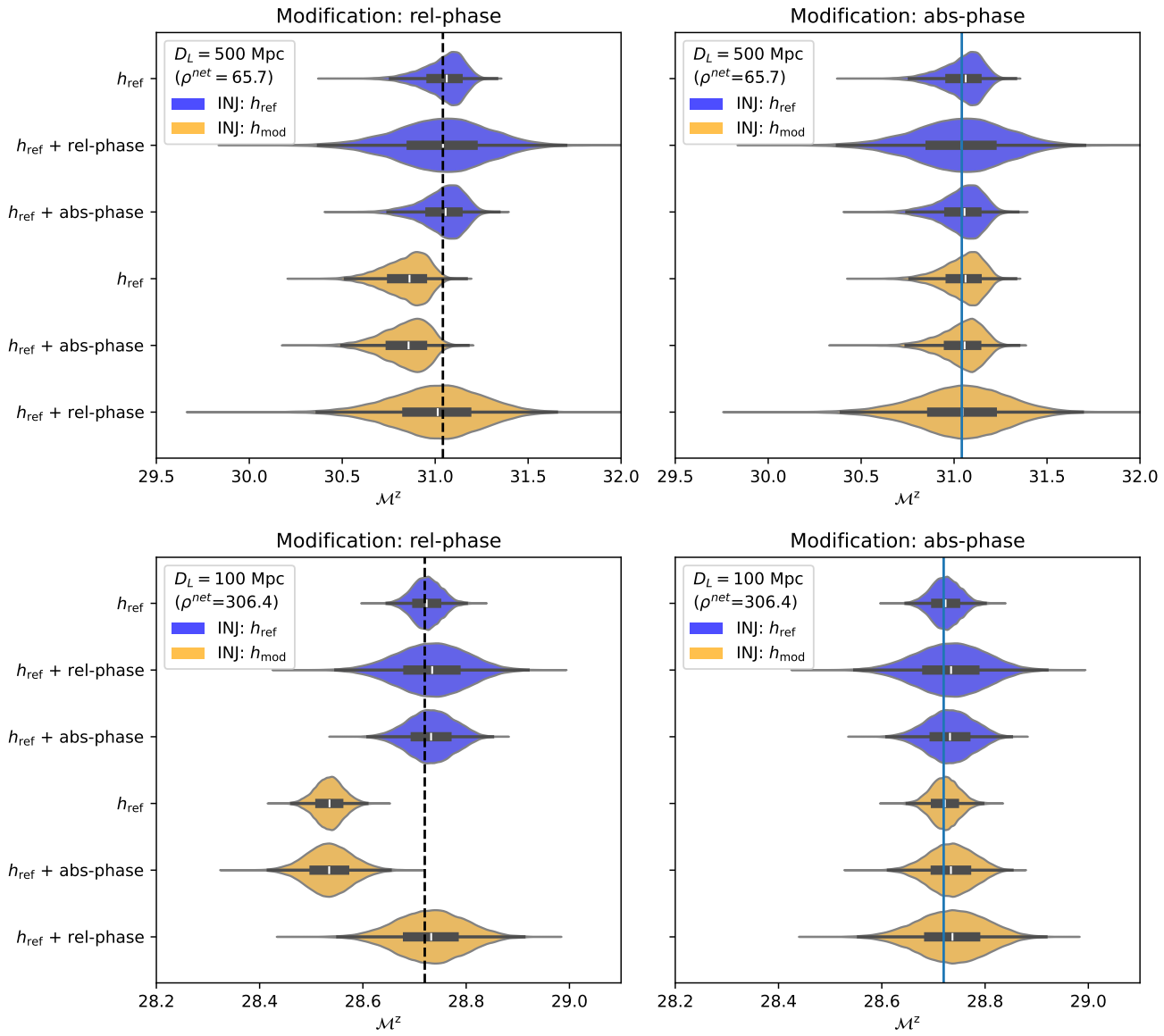


FIG. 9. A GW150914-like signal is injected and recovered using two WF-Error formalisms. The blue violin plots show the cases with reference injection $h_{\rm ref}$ while orange violin plots show the modified injections. The left panels represent the cases where the modification is done with rel-phase modification, while the right panel shows abs-phase modification. The top rows represent the injections with the source placed at a luminosity distance of 500 Mpc, while the bottom panels show the injection corresponds to the same source placed at 100 Mpc. Labels on the y-axis show the waveform models used in PE runs. For these PE runs, we fix the parameters RA, dec, and polarization to the injected values. The network SNR, $\rho^{\rm net}$, for reference injection is shown in each panel.

model in a manner described by the parametrizations in equations (12) and (13), our framework can correct any bias in source parameters, if present. Additionally, a sufficiently loud signal can recover the frequency dependence of WF-Error parameters. We still need to ensure that the priors are sufficiently broad to capture the deviations. Additionally, the selection of knots or nodal points for cubic splines should accurately reflect the true nature of $\delta\tilde{\phi}$ as a function of frequency. In these examples, we use binning in log frequency and use a total of ten nodal points between 20 Hz and 500 Hz.

C. Incorporating missing physics

One use of WF-Error parametrization can be to account for the missing physics. For BBH mergers, the most up-to-date waveform model includes effects such as higher-order harmonics and system precession, and the waveform developers are working on systems with